PSEUDOCODE

PART A

```
IMPORT data from excel
DEFINE permissions to check login details
Create variable username and password
INPUT username from dataset
INPUT password from dataset
CREATE variable user which will check username where on data where condition is true
IF user.empty:
    Display “(username not found”)
ELSE:
     Display (“username found”)
CREATE variable stored_pword
IF password == stored_pword:
  Display ("Login successful")
  CREATE variable role
  DISPLAY (“welcome”, username,” your role is”, role)
IF role in permissions:
        allowed_functions = permissions[role]
        Display ("You can access:")
        FOR function in allowed_functions:
            Display ("-", function)
CREATE variable choice
INPUT function you want to access on choice
     if choice in allowed_functions:
       Display ("Access granted to", choice)
     else:
       Display ("Access denied")
else:
DISPLAY ("Unknown role")
if role == "Admnistrator":
DISPLAY ("Full acces granted")
elif role == "Safety Officer":
DISPLAY ("Access to safety functions granted")
elif role == "Mining Engineer":
DISPLAY ("Access to mining engineering functions granted")
elif role == "Maintenance Engineer":
DISPLAY ("Access to maintenance functions granted")
elif role == "Manager":
DISPLAY ("Access to management functions granted")
else:
DISPLAY ("Unknown role")
else:
DISPLAY ("Incorrect password")
```

PART B

START

LOAD the data Worker Safety from excel.
SHOW the page of Worker Health and Safety.
SHOW the records of all worker safety.
User is allowed to search for a worker using Worker ID.

IF the user selects "Add Worker" THEN
INPUT Worker ID.
INPUT Department.
INPUT PPE Compliance.
INPUT Safety Training Status.
INPUT Fatigue Level.
INPUT Risk Level.

Save the new worker record.
SHOW "Worker added successfully".
ELSE
Display an error message.
END IF
ALLOW user to check the safety conditions of the workers.

IF PPE Compliance is "No" OR
Safety Training is "Incomplete" OR
Fatigue Level is "High" OR
Risk Level is "High" or "Critical" THEN

DISPLAY a safety warning.

ELSE
Show "No safety warnings detected".
END IF
Allow the user to have the access of downloading worker records.
END

PART C

BEGIN Safety Incident Database
LOAD the "Safety Incidents" sheet from the Excel workbook
DISPLAY available actions:
1. Add a new incident
2. View all incidents
3. Search for an incident

```
    4. Filter incidents
    5. Categorise and count incidents
    6. Analyse incident trends
  INPUT user choice
  IF choice = 1 THEN
    INPUT incident details
    GENERATE a unique incident ID
    ADD the incident to the database
    SAVE the updated sheet to the Excel workbook
  ELSE IF choice = 2 THEN
    DISPLAY all incident records
  ELSE IF choice = 3 THEN
    INPUT incident ID
    SEARCH for the incident
    DISPLAY the matching record
    OR DISPLAY "Incident not found"
  ELSE IF choice = 4 THEN
    INPUT filter column and value
    DISPLAY matching incident records
  ELSE IF choice = 5 THEN
    INPUT category: Incident Type or Severity
    GROUP incidents by the selected category
    COUNT incidents in each category
    DISPLAY the counts
  ELSE IF choice = 6 THEN
    CONVERT incident dates to date format
    GROUP incidents by month
    COUNT incidents per month
    DISPLAY the monthly counts
  ELSE
    RETURN to the main application
  END IF
END Safety Incident Database
```

PART D

```
BEGIN Equipment Database
  LOAD the "Equipment" sheet from the Excel workbook
  DISPLAY available actions:
    1. Add equipment
    2. View all equipment
    3. Search for equipment
    4. Filter equipment records
  INPUT user choice
  IF choice = 1 THEN
    INPUT equipment details
```

```
    GENERATE a unique equipment ID
    ADD equipment to the database
    SAVE the updated sheet to the Excel workbook
  ELSE IF choice = 2 THEN
    DISPLAY all equipment records
  ELSE IF choice = 3 THEN
    INPUT equipment ID
    SEARCH for the equipment
    DISPLAY the matching record
    OR DISPLAY "Equipment not found"
  ELSE IF choice = 4 THEN
    INPUT filter column and value
    DISPLAY matching equipment records
  ELSE
    RETURN to the main application
  END IF
END Equipment Database
```

PART E

```
FUNCTION CheckEquipmentCondition ():
  INPUT equipment_id
  INPUT temperature
  INPUT vibration

FUNCTION CheckEquipmentCondition (temperature, vibration):
  IF temperature < 80:
     display "NORMAL: Equipment operating within defined monitoring limits."    ELSE:
        display "WARNING: Equipment temperature approaching predefined threshold."
        display "Inspection required"

  IF vibration < 5:
        display "NORMAL: Equipment operating within defined monitoring limits."    ELSE
IF vibration <= 8:
        display "WARNING: Equipment temperature approaching predefined threshold."
        display "Inspection required"
  ELSE:         display "CRITICAL: Equipment vibration exceeds predefined threshold."
        display "Inspection required"
END FUNCTION
```

PART F

```
FUNCTION MonitorEquipmentMaintenance():
  DISPLAY "EQUIPMENT MAINTENANCE MONITORING "

  INPUT equipment_id
  INPUT operating_time (hours)
  INPUT downtime (hours)
```

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  INPUT hours_since_service
  INPUT maintenance_interval (hours)
  INPUT number_of_alerts

  IF any of the values (except equipment_id) is not a valid number:
     DISPLAY "Error: Enter valid numerical values."
     RETURN
  END IF

  IF operating_time < 0 OR downtime < 0 OR hours_since_service < 0 OR
maintenance_interval <= 0 OR number_of_alerts < 0:
     DISPLAY "Error: Enter valid non-negative values."
     DISPLAY "Maintenance interval must be greater than zero."
     RETURN
  END IF

  INPUT maintenance_status ("yes" or "no")

  IF maintenance_status is not "yes" AND not "no":
     DISPLAY "Error: Enter yes or no."
     RETURN
  END IF

  total_time = operating_time + downtime

  IF total_time > 0:       availability = (operating_time / total_time) * 100
  ELSE:
    availability = 0
  END IF

  DISPLAY "--- MAINTENANCE REPORT ---"
  DISPLAY equipment_id, operating_time, downtime, availability

  IF hours_since_service >= maintenance_interval:
    DISPLAY "STATUS: Maintenance is due."
    IF hours_since_service > maintenance_interval AND maintenance_status is not
"yes":
       DISPLAY "OVERDUE: Maintenance is required."
    END IF
  ELSE:
    DISPLAY "STATUS: Maintenance is not yet due."
  END IF

  IF maintenance_status is "yes":
    DISPLAY "STATUS: Equipment is under maintenance."
  END IF
```

```
  IF total_time > 0 AND (downtime / total_time) >= 0.20:
    DISPLAY "WARNING: Excessive equipment downtime."
  END IF

  IF number_of_alerts >= 3:
    DISPLAY "WARNING: Repeated equipment alerts."
  END IF

  IF availability < 85:
    DISPLAY "WARNING: Low equipment availability."
  END IF

  DISPLAY "Maintenance monitoring completed."
END FUNCTION

CALL MonitorEquipmentMaintenance()
END
```

PART G

BEGIN
DEFINE FUNCTION risk_score
INPUT Likelihood in a scale (1-5) INPUT Consequence in a scale (1-5) Calculate Risk_Score:
Risk_Score = Likelihood*Consequence IF Risk_Score < 0 or Risk_Score >25:
Classification = “Invalid”
PRINT(“Invalid Risk Score”) ELIF Risk_Score <= 4:
Classification = “Low” ELIF Risk_Score <=9:
Classification = “Medium” ELIF Risk_Score <=16:
Classification = “High” ELSE:
Classification = “Critical”
PRINT(“Risk Score:”, Risk_Score, “Classification:”, Classification) IF Classification == “High”:
PRINT(“WARNING! High Risk identified; immediate attention required”) ELIF Classification== “Critical”:
PRINT(“WARNING! Critical Risk identified; immediate attention required”)
END FUNCTION
CALL FUNCTION risk_score
END

PART H

BEGIN
DEFINE FUNCTION monitoring_alerts
INPUT Equipment ID

```
IF Equipment ID does not exist THEN
  DISPLAY "Equipment ID not found"
ELSE
  DISPLAY Equipment ID and Equipment Type
  IF Temperature > 100 THEN
    DISPLAY "CRITICAL: High temperature"
  ELSE IF Temperature >= 80 THEN
    DISPLAY "WARNING: High temperature"
  END IF
  IF Vibration > 8 THEN
    DISPLAY "CRITICAL: High vibration"
  ELSE IF Vibration >= 5 THEN
    DISPLAY "WARNING: High vibration"
  END IF
  IF Downtime > 24 THEN
    DISPLAY "WARNING: Excessive equipment downtime"
  END IF
  IF Maintenance Status = "Due" OR "Overdue" THEN
    DISPLAY "WARNING: Maintenance due or overdue"
  END IF

  IF Critical Safety Incidents are present THEN
    DISPLAY "NOTICE: Critical safety incidents recorded"
  END IF
  IF High-Risk Safety Observations are observed THEN
    DISPLAY "NOTICE: High-risk safety observations recorded"
  END IF
  IF High or Critical Risk Assessments are present THEN
    DISPLAY "NOTICE: High or critical risks recorded"
  END IF
  IF No alerts were generated THEN
    DISPLAY "No alerts, all conditions normal"
  END IF
END IF
CALL FUNCTION monitoring_alerts
END
```

PART I

```
Load the data from the dataset
  IF the tables cannot be loaded
    DISPLAY a data-loading error
    STOP
  END IF

  SET workers = Worker Safety records
  SET incidents = Safety Incidents records
  SET observations = Safety Observations records
```

SET equipment = Equipment records
SET alerts = Alerts records

CALCULATE total_workers
AS the number of unique Worker IDs

CALCULATE total_incidents
AS the number of unique Incident IDs

CALCULATE near_misses
AS the number of incidents whose Incident Type is "Near Miss"

CALCULATE high_risk_observations
AS the number of observations whose Risk Level is "High" or "Critical"

CALCULATE ppe_compliance
AS the percentage of workers whose PPE Compliance is "Yes"

CALCULATE total_equipment
AS the number of unique Equipment IDs

CALCULATE available_equipment AS the number of unique Equipment IDs whose Maintenance Status is not "Under Maintenance"

CALCULATE equipment_under_maintenance
AS the number of equipment records whose Maintenance Status is "Under Maintenance"

SELECT equipment_alerts
FROM alerts where Source is "Equipment"

CALCULATE total_equipment_alerts
AS the number of equipment alert records

CALCULATE average_equipment_availability
AS the mean of Availability (%) in equipment records

SELECT critical_open_alerts
FROM equipment_alerts where Severity is "Critical"
AND Status is "Open"

CALCULATE critical_equipment_alerts
AS the number of critical_open_alerts

DISPLAY all calculated dashboard indicators

END PART I

PART J

```
BEGIN

  RECEIVE the application tables

  DISPLAY the available analysis questions
  GET the question selected by the user

  IF the selected question is about incidents by department
    GET a date range
    FILTER incidents to the selected date range
    COUNT unique Incident IDs for each Department
    SORT department counts from highest to lowest

    IF no incident counts are available
      DISPLAY "No incidents in the selected period"
    ELSE
      DISPLAY the department with the highest count
      DISPLAY a bar chart and the department-count table
    END IF

  ELSE IF the question is about highest downtime or lowest availability
    GET the selected measure: Downtime or Availability
    SORT equipment by that measure

    IF no equipment records are available
      DISPLAY an information message
    ELSE
      DISPLAY the equipment with the highest downtime
      OR the equipment with the lowest availability
      DISPLAY a chart and a table of up to ten equipment records
    END IF

  ELSE IF the question is about highest vibration or temperature
    GET the selected measurement
CONVERT its values to numeric form
    REMOVE records without a usable value
    SORT equipment from highest reading to lowest

    IF no readings are available
      DISPLAY an information message
    ELSE
      DISPLAY the equipment with the highest reading
      DISPLAY a chart and a table of the ranked equipment
```

```
  END IF

ELSE IF the question is about equipment alerts
  SELECT alert records whose Source is "Equipment"
  COUNT unique Alert IDs for each equipment Record ID
  SORT alert counts from highest to lowest

  IF no equipment alerts are available
    DISPLAY an information message
  ELSE
    DISPLAY the equipment with the highest alert count
    DISPLAY a chart and an alert-count table
  END IF

ELSE IF the question is about maintenance priority
  CALCULATE maintenance indicators for each equipment record
  IDENTIFY records with maintenance flags
  SORT flagged equipment by the priority indicators
  DISPLAY the flagged equipment and its maintenance details

  IF no equipment is flagged
    DISPLAY a message that no equipment matches the conditions END IF

ELSE IF the question is about common incident causes
  COUNT incident records for each Cause
  DISPLAY a chart and a cause-count table

ELSE IF the question is about high-risk observations by department
  SELECT observations with Risk Level "High" or "Critical"
  COUNT unique Observation IDs for each Department
  SORT counts from highest to lowest
  DISPLAY a chart and a department-count table

ELSE IF the question is about departments with safety    and equipment concerns
  GET a date range for incidents
  FILTER incidents to that date range

  GET a date range for safety observations
  FILTER observations to that date range

  BY Department, CALCULATE:
    Number of safety incidents
    Number of high or critical incidents
    Number of high or critical observations
    Average equipment downtime
    Number of recorded alerts
```

COMBINE the departmental results
REPLACE missing department values with zero SORT the comparison by high or critical incidents, high or critical observations, and average downtime DISPLAY the departmental comparison table

ELSE
ASSESS temperature and vibration condition for each equipment item
COUNT equipment items for which both readings are classified Normal
CALCULATE the percentage of such equipment items
DISPLAY the count, percentage, and condition results
END IF

END PART J